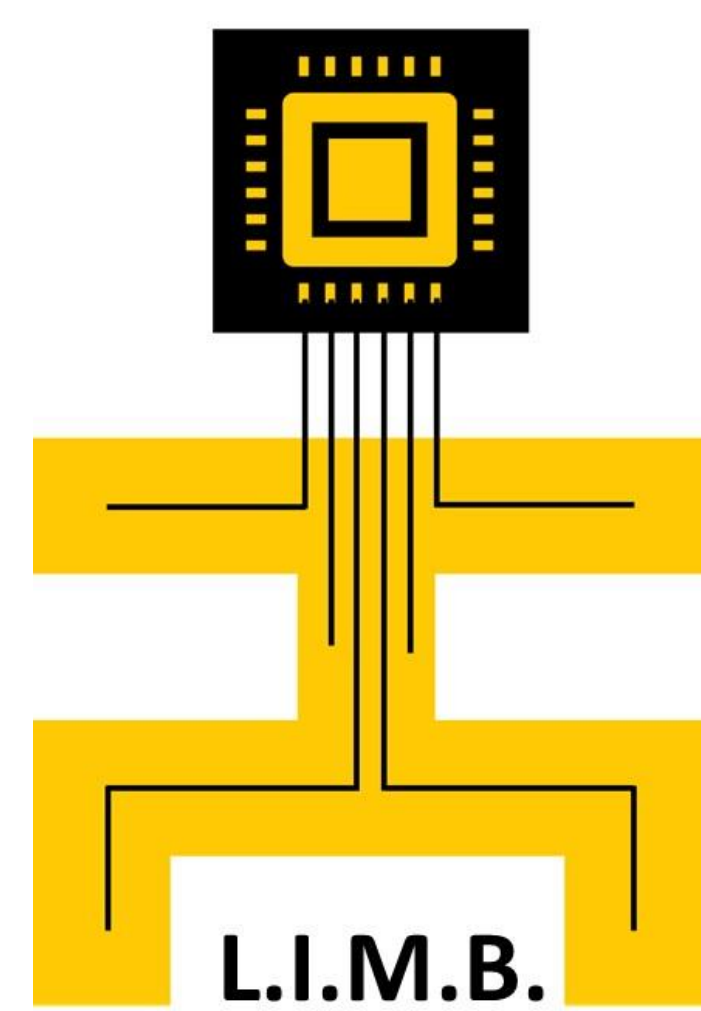




Laboratory for
Interaction of
Machine and
Brain

Prosthetic Hand Force Control via sEMG-Derived Intent

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Introduction

sEMG is a measure of muscle activity used by machine learning models to infer user intent for prosthetic hand control



Fig. 1a: sEMG open



Fig. 1b: sEMG pinch

Many users report difficulty performing everyday grasping tasks with current control schemes [2]

Studies report prosthesis abandonment rates of up to 40–50% [2]



Fig. 2a: Over-grip

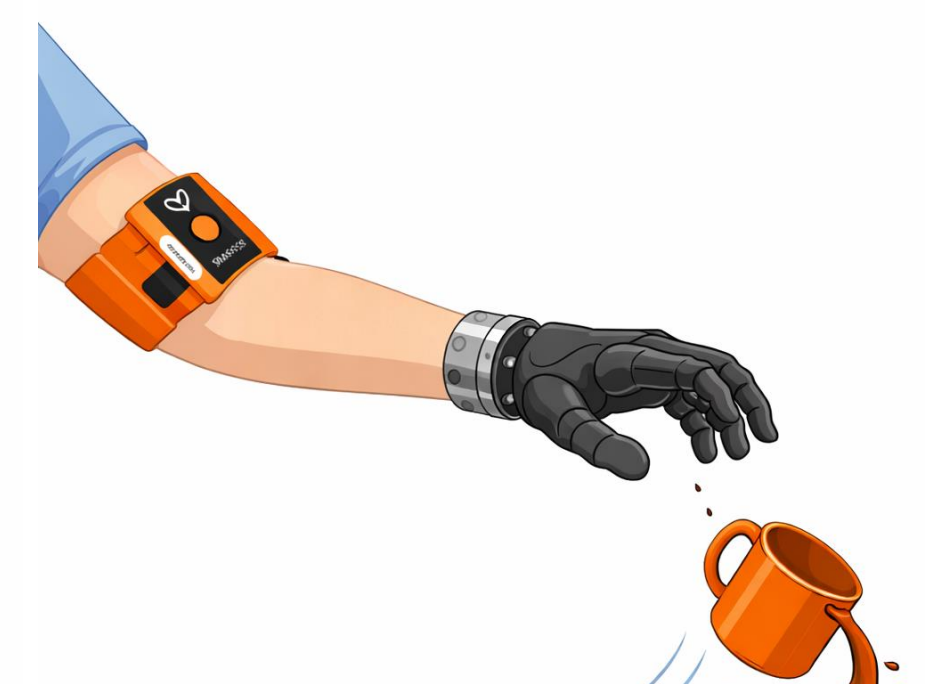


Fig. 2b: Under-grip

Hypothesis

Object grasping becomes more intuitive when grasp force regulation is handled by the prosthetic hand

Methods

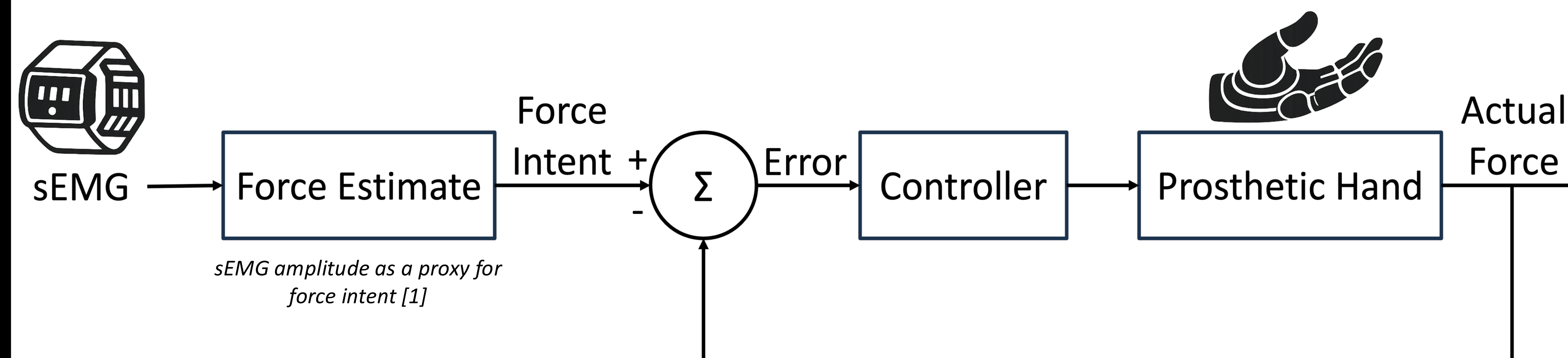


Fig. 3: Algorithm diagram

The algorithm is triggered upon object contact, shifting force regulation to the prosthetic hand

Results

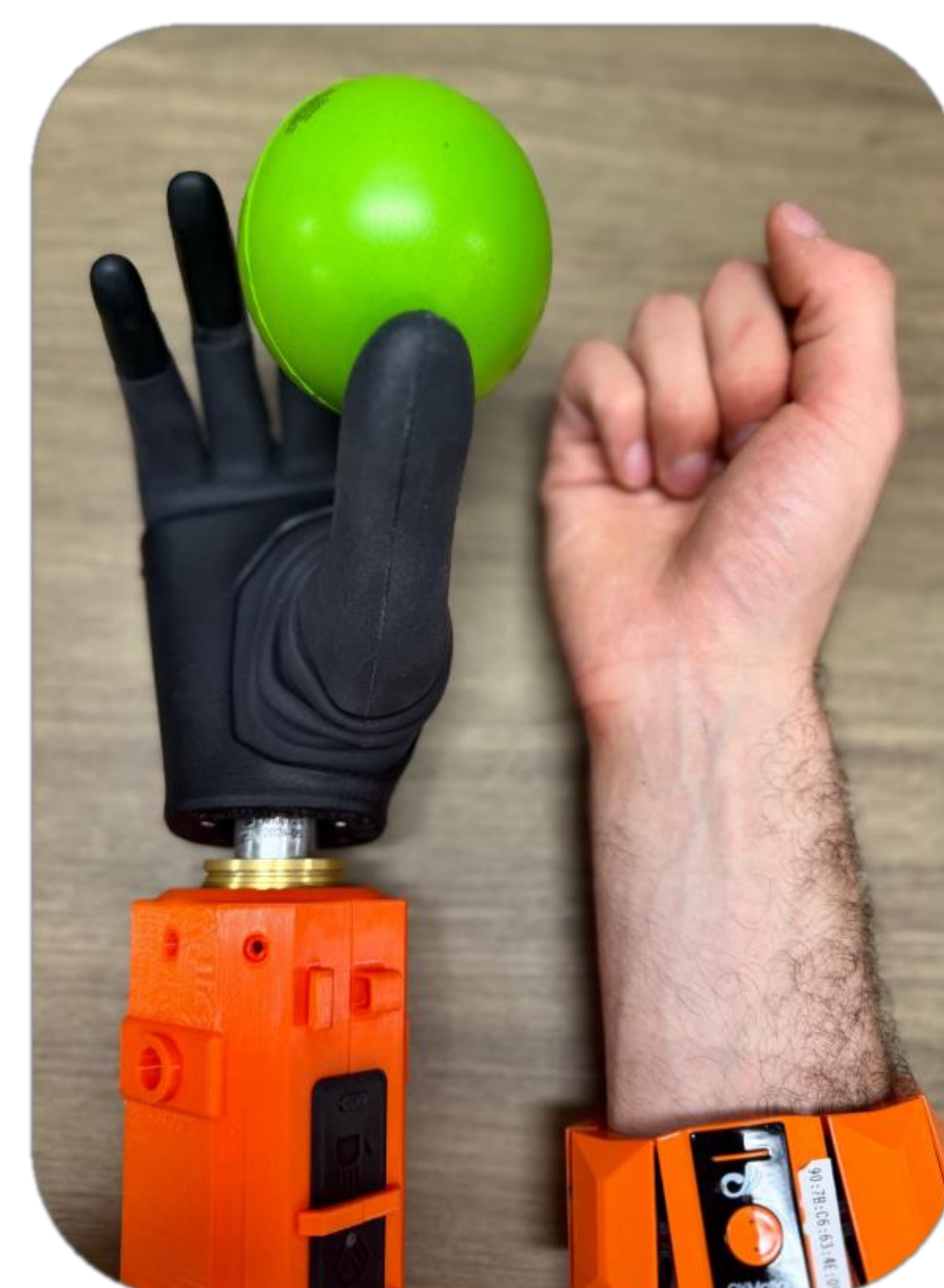


Fig. 4a: Low grip force



Fig. 4b: High grip force

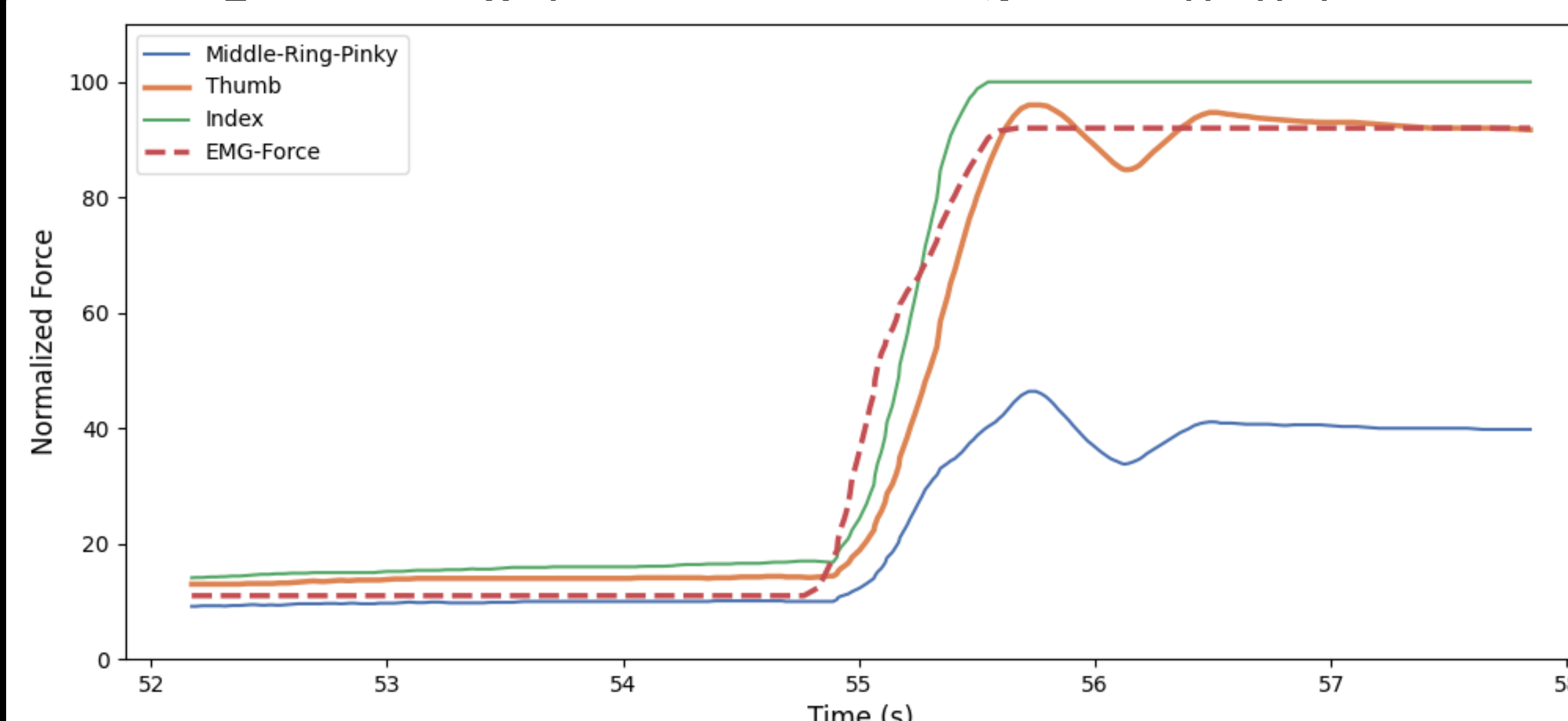


Fig. 4c: Force sensor readings during grip increase

Demo



Fig. 5: Firm Object Grasp



Fig. 6: 15 Second Demo
Video

Limitations

User force estimation was evaluated under controlled conditions

Further evaluation is required across a wider range everyday objects

Conclusion

sEMG estimates user's force intent while the prosthetic hand regulates it

Results show the prosthetic hand adjust grip force to match user intent

This enables more natural, biologically aligned approach to hand control



Fig. 7: Adaptive grip

References

- [1] De Luca, C. J. (1997). The use of surface electromyography in biomechanics. *Journal of Applied Biomechanics*, 13(2), 135–163.
- [2] Biddiss, E. A., & Chau, T. T. (2007). *Upper limb prosthesis use and abandonment: A survey of the last 25 years*. *Prosthetics and Orthotics International*, 31(3), 236–257.